

■ 3.6.2 Advanced Topic: The Representation of Power Series

Power series are represented in *Mathematica* as `SeriesData` objects.

The power series is printed out as a sum of terms, ending with `0[x]` raised to a power.

```
In[1]:= Series[Cos[x], {x, 0, 4}]
```

$$\text{Out}[1]= 1 - \frac{x^2}{2} + \frac{x^4}{24} + 0[x]^5$$

Internally, however, the series is stored as a `SeriesData` object.

```
In[2]:= InputForm[%]
```

```
Out[2]//InputForm=
SeriesData[x, 0, {1, 0, -1/2, 0, 1/24}, 0, 5, 1]
```

By using `SeriesData` objects, rather than ordinary expressions, to represent power series, *Mathematica* can keep track of the order and expansion point, and do operations on the power series appropriately. You should not normally need to know the internal structure of `SeriesData` objects.

You can recognize a power series that is printed out in standard output form by the presence of an `0[x]` term. This term mimics the standard mathematical notation $O(x)$, and represents omitted terms of order x . For various reasons of consistency, *Mathematica* uses the notation `0[x]^n` for omitted terms of order x^n , corresponding to the mathematical notation $O(x)^n$, rather than the slightly more familiar, though equivalent, form $O(x^n)$.

Any time that an object like `0[x]` appears in a sum of terms, *Mathematica* will in fact convert the whole sum into a power series.

The presence of `0[x]` makes *Mathematica* convert the whole sum to a power series.

```
In[3]:= a x + Exp[x] + 0[x]^3
```

$$\text{Out}[3]= 1 + (1 + a) x + \frac{x^2}{2} + 0[x]^3$$